

Mahathir Mohammad Shuvo

Full-Stack Software Engineer

Dhaka, Bangladesh | shuvo1728@gmail.com | +880 1743 078887 | linkedin.com/in/mahathir-mohammad-shuvo | github.com/MahathirMohammadShuvo

SUMMARY

Full-stack engineer with six years of professional experience shipping production systems across web, mobile and desktop. Currently develops four concurrent production codebases in TypeScript, Go, Elixir and Python: REST and WebSocket backends, React and React Native front ends, a Flutter client, an Electron desktop app and distributed Go services. Owns systems end to end - schema and API design, real-time features, payments, CI to the app stores, automated testing and production monitoring. Open-source contributor to PDFKit and Radix UI.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, Go, Elixir, Dart, PHP, SQL
Backend: NestJS, Node.js, Express, Fastify, Flask, Phoenix, Prisma, REST APIs, WebSockets, Socket.IO, Zod
Databases: PostgreSQL, MySQL, MongoDB, Redis, Elasticsearch, SQLAlchemy, LevelDB
Frontend: React, TanStack Router/Query/Table/Form, Redux, Zustand, Tailwind CSS, Radix UI, Vite
Mobile / Desktop: React Native (Expo), Flutter, Electron, Codemagic CI, EAS Build
AI: Model Context Protocol (MCP), LangGraph, Anthropic API, RAG, Hugging Face Transformers, TensorFlow.js
Cloud / DevOps: AWS, Google Cloud Platform, Docker, nginx, Apache, PM2, GitHub Actions, Vercel, Netlify
Security / Testing: Ed25519, X.509, WebCrypto, JWT, OAuth, bcrypt, Playwright, Vitest, Jest, Mocha, Patrol
Other: WebRTC (mediasoup, LiveKit), web scraping, Google Maps API, Stripe, PDF and Excel generation

EXPERIENCE

Software Engineer

Apr 2024 - Present

Confidential client under NDA - Remote

- Develop a large TypeScript monorepo spanning HTTP backend, web app, admin console, mobile and desktop clients, with roughly a quarter of the codebase being automated tests.
- Build distributed Go services on LevelDB, including Ed25519 signing/verification and canonical serialisation between nodes.
- Ship a cross-platform Electron desktop app with signed auto-updates and PDF processing; integrate biometric identity verification (AWS Rekognition, Amplify Liveness, TensorFlow.js).
- Implement metered billing, subscriptions and payouts with Stripe, plus a rate-limited public API with key management.

Full-Stack Developer

Jan 2021 - Mar 2024

DKERPS - Remote

- Sole backend developer across five products, owning deployment, testing and architecture on several concurrently.
- Built a Flask/SQLAlchemy large-file transfer service supporting single transfers up to 40 GB, on GCP with Ubuntu and Apache.
- Built Dropbox-style cloud storage with shared folders, business accounts and per-member permissions, on AWS with nginx at 99% uptime; and a centralised log-management system on Elasticsearch.
- Built a Zoom-style conferencing app (MERN, mediasoup WebRTC) and delivered a React Native (Expo) scheduling app - 100% of backend, 70% of frontend.

PROJECTS

- Discount Bajar (discount-bajar.com) - live e-commerce platform. Four apps on one NestJS/Prisma/PostgreSQL backend: React web, admin console, Expo app for iOS and Android, and workers. Double-entry accounting with bank reconciliation, rider dispatch, referral and reward engine, recommendation engine, RAG product search, real-user monitoring.
- AI Employees - self-hosted multi-agent platform. Agents organised as a company collaborating over the Model Context Protocol; LangGraph orchestration with a PostgreSQL checkpointer, Redis-backed Socket.IO, React UI with Monaco editor and node-graph monitoring.
- Daanika - real-time multiplayer game platform. Flutter client with 11 board games on an Elixir/Phoenix backend; matchmaking, tournaments and live spectating over Phoenix channels, WebRTC camera streaming via LiveKit, native end-to-end tests with Patrol.

OPEN SOURCE

- PDFKit (10.7k stars) - two merged pull requests: AcroForm date fields emitting invalid JavaScript so formatting never applied, and indentAllLines applying the indent repeatedly across paragraphs and page breaks.
- Radix UI (19.2k stars) - pull request fixing DismissibleLayer requiring two outside clicks to close after inner content stopped event propagation.

Languages: Bengali (native), English (professional), Hindi (conversational)